

OPENDESIGN

OpenDesign — HelixCode Integration

Overview

OpenDesign (<https://github.com/nexu-io/open-design>) is a local-first, open-source design workspace that turns coding agents into design engines. It provides design artifact generation, design systems, and design tokens.

Constitution reference: §11.4.162 (OpenDesign UI design system mandate)

Components

1. **OD Daemon** — Express + SQLite local server (port 7456)
2. **open-design-mcp** — MCP stdio server bridging agents to the daemon
3. **Design systems** — 150+ DESIGN.md brand files
4. **Skills** — 100+ design prompts/templates

Installation

MCP Server (installed)

```
npm install -g open-design-mcp  
# Version: 0.16.1
```

Daemon (requires setup)

The daemon can be run via Docker or from source:

Docker:

```
git clone git@github.com:nexu-io/open-design.git /path/to/open-  
design  
cd /path/to/open-design/deploy  
cp .env.example .env  
docker compose up -d
```

From source:

```
git clone git@github.com:nexu-io/open-design.git
cd open-design
corepack enable && pnpm install
pnpm tools-dev run web
```

MCP Integration

Add to `.mcp.json`:

```
{
  "mcpServers": {
    "open-design": {
      "command": "npx",
      "args": ["-y", "open-design-mcp"],
      "env": {
        "OD_DAEMON_URL": "http://localhost:7456"
      }
    }
  }
}
```

Available MCP Tools

Tool	Description
od_list_projects	List all design projects
od_get_project	Fetch project + artifact files
od_create_project	Create new design project
od_save_artifact	Save HTML artifact to global store
od_save_project_file	Save file inside a project
od_lint_artifact	Lint HTML for issues
od_compose_brief	Format design prompt
od_generate_design	Generate design via BYOK (requires API key)

Design Token Integration (§11.4.162)

Per constitution §11.4.162: - Use OpenDesign's design tokens/themes for color palette (light + dark) - Typography scale and spacing - Component-level design tokens - Project brand colors from canonical assets (assets/Logo.png)

BYOK Configuration (for design generation)

To enable `od_generate_design`, add BYOK vars:

```
{
  "env": {
```

```

"OD_DAEMON_URL": "http://localhost:7456",
"BYOK_BASE_URL": "https://your-ai-proxy.example.com/v1",
"BYOK_API_KEY": "<provider-api-key>",
"BYOK_MODEL": "open-design",
"BYOK_PROVIDER": "openai"
}
}

```

Cross-references

- constitution/Constitution.md §11.4.162 (OpenDesign UI mandate)
- constitution/Constitution.md §11.4.74 (extend-don't-reimplement)
- constitution/Constitution.md §11.4.35 (project-specific configuration)

Sources verified 2026-06-22: <https://github.com/nexu-io/open-design> , <https://registry.npmjs.org/open-design-mcp/latest>

Cross-referenced this doc against the latest official OpenDesign sources (fetched 2026-06-22). Findings: - **MCP package version current.** npm open-design-mcp latest = **0.16.1** (registry latest tag; description “MCP stdio server bridging coding agents to Open Design daemon (BYOK flow with full systemPrompt fidelity).”). This doc’s “open-design-mcp ... Version: 0.16.1” claim is accurate. - **Components + daemon port confirmed.** The upstream README confirms the two components this doc lists: an Express + SQLite **daemon** on port **7456** (<http://localhost:7456>) and a **stdio MCP server**. It also corroborates the 150+ design-system DESIGN.md brand files and 100+ skills this doc cites (upstream states “100+ Skills”, “150+ Design Systems”, “261 Plugins”). - **Negative finding (install method drift).** The upstream now documents a one-line installer — `curl -fsSL https://open-design.ai/install.sh | sh -s <agent>` — and from-source guidance of **Node ~24 + pnpm 10.33.x** (this doc says pnpm install). The doc’s `npm install -g open-design-mcp` still yields the published 0.16.1 MCP server, so it is not wrong, but the upstream curl-installer is the now-preferred path and the from-source toolchain pins (Node 24 / pnpm 10.33.x) should be reflected on the next revision.